Study Information

SURVEY OF UNDERGRADUATE TEACHING OF QUANTITATIVE STUDY DESIGN AND ANALYSIS FOR LIFE SCIENCES ACROSS THE UK

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About a decade ago, evidence emerged that results from published scientific studies were less "reproducible" than desired, and thereby cast doubt on scientific integrity (Begley & Ellis, 2012; Baker 2016). Since then, accumulating evidence has indicated that a reproducibility crisis spans multiple scientific disciplines (e.g., psychology (Open Science Collaboration 2015); biomedical science (Freeman et al. 2015). Low reproducibility presents serious ethical challenges for scientific progress.

What factors underlie poor research reproducibility? Inadequate experimental design plays a role. For example, low statistical power combined with publication bias (favouring "statistically significant results") inflate Type 1 errors (Ioannidis 2005; Button et al. 2013; Parker et al 2016). Inappropriate statistical analyses also reduce reproducibility. For example, analyses of animal studies frequently fail to account for non-independent observations that increase Type 1 error rates (Lazic et al 2018). In addition to technical problems with study design and analysis, researchers can frequently make decisions during data collection and analysis that endanger reproducibility (so called "Questionable Research Practices"; e.g., p-hacking; Hypothesizing After Results are Known ("HARKing"; Kerr 1998); John et al 2012; Fraser et al 2018).
These factors that lower reproducibility may exist for multiple reasons. In some cases, researchers will adopt poor practice to increase chances of publication and gain the associated rewards, e.g., in extreme cases, fabricating data (Fraser et al 2018). We suspect, however, that such behaviours occur rarely and, more commonly, researchers employ poor practice due to insufficient training. This training begins at undergraduate level. However, few data exist that describe student training in study design and analysis to assess how they may impact research reproducibility across disciplines.

This project aims to characterize undergraduate training in study design and analysis for four life science disciplines: Psychology, Biology, Biomedical Science and Medicine. We will use a survey to characterize training in (quantitative) study design and analysis. We focus on these four disciplines because, despite having varied research foci and employing different forms of measurements (data), they frequently adopt comparable experimental design that call for similar analyses. We hope that these similarities will provide opportunities for disciplines to learn from one another to improve undergraduate training, where discrepancies exist among disciplines.

For the purposes of our survey, we define the four disciplines as follows:
Psychology: Undergraduate degree in psychology accredited by the British Psychological Society (BPS);
Biomedical Sciences: The study of biology with a long-term focus to improve human and animal health by improving understanding that benefits healthcare (e.g., Medical Science) or public health (e.g., Neuroscience, Reproductive Biology, Pharmacology, Infectious Diseases)
Biological Sciences: Biology, broadly defined, is the study of life. However, for the purposes of our survey, we distinguish "biology" from "biomedical sciences". Hence, we restrict our definition of Biological Sciences to the study of biology without a long-term focus of improving human or animal health (e.g., Ecology, Evolution, Plant Sciences, Molecular Biology).
Medicine: a medical degree accredited by the General Medical Council.

Further, we note student sub-disciplines, where applicable. We aim to document teaching for (largely) independent cohorts of undergraduate students; i.e., our analysis uses 'student cohort' as a unit of observation (data point). We recognize, however, that sometimes multiple student cohorts (i.e., sub-disciplines) receive training independently within a Discipline at a University. Therefore, to analyse the survey results appropriately, we require survey respondents to indicate the sub-discipline (i.e., programme/specialisation) to which the survey answers apply, if applicable. This means that we may ask for the survey to be completed multiple times (once per sub-discipline) for a given Discipline at a University. In such cases, we are happy for different people to complete separate instances of the survey for various sub-Disciplines at a given university.

We obtained Ethical Approval for our research from the School of Philosophy, Psychology and Language Sciences at the University of Edinburgh (ref 326-2223/1).

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Hypotheses

Our study is descriptive and exploratory in nature and, with two exceptions (below), we do not test *a priori* hypotheses. Specifically, we address these three aims using **quantitative results:** (1) identify topics that are vs. are not taught universally across (sub)disciplines in order to highlight areas where additional training may be required, universally; (2) identify areas where (sub)disciplines tend to differ in the amount (or probability) of training on a given topic and thereby point to areas where disciplines may learn from one another (3) characterize and contrast teaching approaches among (sub)disciplines (e.g., using devoted courses vs. embedded in other courses; multiple vs. single year / episodes of teaching) to identify opportunities for (sub)disciplines to learn from each other to improve undergraduate training. Our survey also opportunistically collects **qualitative data** (i.e., open response questions) to collate information that may identify effective teaching practice.

As a minor component of our study, we address two *a priori* hypotheses that:

(i) Within Biological Sciences, programmes (at Discipline or sub-discipline level) that teach ecology and/or evolution devote more time (credits) to training undergraduates in study design and/or analysis than those Biological Science programmes that do not teach ecology or evolution. We base this hypothesis both on (anecdotal) personal experience, and for historical reasons: e.g., Ronald Fisher, who contributed much understanding to evolutionary biology contributed similarly to statistics; e.g., ecology and evolution have long histories of mathematical (theoretical) studies.

(ii) The discipline of Psychology tends to devote more time (credits) on (a) quantitative study design and b) data analysis than the remaining three disciplines. We base this hypothesis on anecdotal evidence and aim to test it formally.

Design Plan

Study Type

Observational study - Data is collected via a survey, from study subjects that are not randomly assigned to a treatment.

Blinding

No blinding is involved in this study.

Is there any additional blinding in this study?

N/A

Study Design

Our study administers a pre-defined survey to qualified UK university staff. Since the previous pre-registration (22.06.2023), we have made several changes to the survey (attached). Questions 5b(i) and 5c(i) now have 10 possible responses instead of 9. Where applicable, our 7 survey respondents will be contacted to amend their answers for this question. We have also made minor edits to the survey to improve clarity.

Sampling Plan

Existing Data

Registration following analysis of the data: As of the date of submission, you have accessed and analyzed some of the data relevant to the research plan. This includes preliminary analysis of variables, calculation of descriptive statistics, and observation of data distributions. Please see cos.io/prereg for more information.

We have received 7 responses to the survey, which highlighted a few minor issues in the survey. For example, we had not accounted for Medicine when asking how many years the degree is, so a participant from this discipline had to indicate the number of years using our "Other" comment box. We have since amended the drop-down box to include “5” and “6” as possible answers. This change is intended to improve clarity only.

We have looked at the data distributions (i.e. histograms) of our 7 responses, which allowed us to talk about the study at a conference. Please see our previous pre-registration (22.06.2023) for details provided prior to data collection.

Data Collection

We are recruiting people who have knowledge about undergraduate study design and data analysis teaching for a discipline in the life sciences at their institution. For a person to be contacted, their knowledge must be for a discipline in the life sciences which matches one of our four definitions. For example, we specify that psychology degrees must be BPS-accredited in

order to be eligible in the study. We aim to contact one person per discipline per institution, so our first point of contact will be someone who has knowledge of study design/data analysis teaching across the entire degree. If this is not possible, we may contact multiple people for a given discipline at a given university, who can work together to complete the survey. Participants will not be paid for their participation.

We expect that the response rate to our survey will increase when respondents are contacted by people familiar to them or from their own institution. Therefore, our survey allows for two approaches to administering the survey: (1) we contact respondents ourselves; (2) we allow volunteers to contact colleagues known to them or at their own institution to recruit respondents for the survey. For example, we plan to use contacts within the UK Reproducibility Network (UKRN) to help with the second sampling approach.

Our sampling will involve several stages. (1) We will begin our sampling by contacting potential respondents (in all four disciplines) at UK universities that have GMC-accredited Medicine programs. As Medicine programs are least common among the four disciplines, this approach maximizes the opportunity to sample all four disciplines at similar universities. We recognize that this set of universities will be a biased sample of UK universities; however, we do not anticipate that this sampling approach will bias **differences among disciplines**, a primary focus of our study (2) Simultaneous to *(1)*, we will accept help from volunteers (e.g. UKRN, members of the research team, and survey respondents) that recruit respondents opportunistically. (3) Among disciplines and universities that remain unsampled in *(1)* and *(2)*, we will select universities (and disciplines therein) at random for contact.

The following text provides details for how the above-mentioned Stage 3 will be completed, by creating a randomised list of eligible universities.

Step 1: Creating a list of eligible universities

For each discipline, we have created a list of universities which offer an undergraduate degree in that discipline (by our definition).

- **Psychology:** universities which offer BPS-accredited degrees, as listed on the BPS website [https://portal.bps.org.uk/Accredited-Courses].
- **Medicine:** universities which offer GMC-accredited degrees, as listed on the GMC website [https://www.gmc-uk.org/education/how-we-quality-assure-medical-education-and-training/evidence-data-and-intelligence/awarding-bodies-and-new-schools].
- **Biomedical Sciences:** universities listed in student enrolment data published by HESA or in the biomedical sciences rankings published by the Complete University Guide, and which meet our criteria.
- **Biological Sciences:** universities listed in student enrolment data published by HESA or in the biological sciences rankings published by the Complete University Guide, and which meet our criteria.

Method for creating a list of eligible universities:

The list of eligible universities in the biological and biomedical sciences is mainly based on data published by the Higher Education Statistics Agency (HESA; https://www.hesa.ac.uk/data-and-analysis/students/what-study), which describes student enrolments in the life sciences for every UK university in the 2020/21 academic year. This was chosen as the most comprehensive by-subject list of UK universities available.

HESA use their own groupings to classify subjects taught in the UK. These groupings do not always match our definitions, particularly across biomedical sciences and medicine, so data for all 4 disciplines was downloaded and then rearranged to meet our criteria.

1. At the CAH level 3 grouping level, student enrolment data were downloaded for each discipline:
    a. Medicine and dentistry
    b. Subjects allied to medicine
    c. Biology and sport sciences
    d. Psychology
2. When downloading, “first degree” was selected to include bachelor degrees only. Data was downloaded for the academic year 2021/2022, as the most recent one available.
3. HESA lists student enrolments by discipline across 244 universities.
4. **PSYCHOLOGY (124 universities)**
    a. Universities with zero student enrolments in psychology were removed from the list, giving a total of 129 universities.
    b. A list of BPS-accredited psychology degrees (from the BPS website) was compared against the HESA list of psychology degrees.
    c. Universities which were not BPS-accredited (or were international) were removed from the list, giving a total of 124 universities.
    d. NOTE: although the list of BPS-accredited universities is sufficient for identifying eligible universities, comparing against the HESA list means that university names are consistent across disciplines.
5. **BIOMEDICAL SCIENCES (116 universities)**
    a. List of subject areas removed from Biomedical Sciences (as a result of not having a substantial biology component):
        i. Nursing and midwifery
        ii. Complementary and alternative medicine
        iii. Counselling, psychotherapy, and occupational therapy.
    b. List of subjects areas added to Biomedical Sciences:
        i. Medical Sciences
    c. Universities with zero student enrolments in the biomedical sciences were removed from the list, giving a total of 116 universities.
6. **BIOLOGICAL SCIENCES (110)**
    a. List of subject areas removed from biological sciences:
        i. Hair and beauty sciences
        ii. Sport and exercise science
    b. Universities with zero student enrolments in the biological sciences were removed from the list, giving a total of 110 universities.
7. A separate list of universities offering Biological Sciences/Biomedical Sciences was sourced from by-subject university rankings published by the Complete University

Guide (2024). The university rankings offer a less comprehensive but more up-to-date list of universities offering that subject. As such, they were used to highlight universities which had been missed in the HESA data. Based on this information, two universities were added to the list of eligible universities offering degrees in the biomedical sciences and none were added to the biological sciences list of eligible universities.

   a. Biomedical Sciences: https://www.thecompleteuniversityguide.co.uk/league-tables/rankings/biomedical-sciences
   b. Biological Sciences: https://www.thecompleteuniversityguide.co.uk/league-tables/rankings/biological-sciences

For all 4 disciplines, international universities were removed from the list of eligible universities (as this is a study focusing on UK universities).

Randomising the list of eligible universities:

1. We compiled and randomised a list of universities which offer either psychology, biological sciences, or biomedical sciences (see above for details).
2. When deciding who to contact, we will approach this list one university at a time. For each university, we will attempt to contact at least one member of staff in psychology, biological sciences, and biomedical sciences. We anticipate some overlap in what is considered biological vs. biomedical science, therefore we will decide on a case-by-case basis whether each university offers biological or biomedical science (even if this was not indicated in the lists created above).
3. In the case of psychology, only universities indicated in the list of eligible universities for that discipline will be contacted, so as to fulfil the criterion that psychology degrees must be BPS-accredited.

Step 2: Procedure for determining who to contact

For each discipline and university, we will use publicly-available information such as the course website to determine who the best person to contact is. Priority will be given to finding one person who is likely to have knowledge about the entire degree course.

If it is not possible to find one person who is able to answer the survey or if it is not clear who to contact, we will send an initial enquiry to several people with knowledge of the course and ask their advice on who to contact. If appropriate, we may ask several people in one discipline at a given university to answer the survey, so that their combined knowledge can be collected. In some instances, it may be appropriate for one person to fill in the survey multiple times, for different disciplines.

In the event that the people outlined above are not able to help (or do not respond), we may ask other members of that department in that university for advice on who to contact. We will move onto the next university once a reasonable effort has been made to contact the relevant people within a discipline at a given university.

Sample size

We aim to survey all (sub-)disciplines at all UK universities that meet the inclusion criteria listed above. Such complete sampling would provide the least biased picture possible of training in study design and analysis for the four disciplines across the UK. However, we may not achieve this complete sampling due to (1) inherent low response rates to surveys (e.g., a colleague reported response rates ~15% for another study); (2) limited time to contact all possible respondents (e.g., we have funding for a Research Assistant until November 2023, at the time of submitting this pre-registration).

At a minimum, we aim to survey 20 UK Universities per discipline. Beyond this, we aim to survey as many (sub-)disciplines at UK universities as permitted within our funding (time) constraints.

If we are contacted by volunteers willing to sample outside of the UK, we will (1) obtain ethical approval for this sampling and (2) allow opportunistic sampling of this sort. We anticipate that any analyses of data from outside the UK will be analyzed separately from UK responses, but will follow the same approach as outlined for UK data.

Stopping rule
We will complete sampling either when all disciplines (at all UK Universities) have been sampled, or when limited time and resources preclude further sampling. Within time constraints, all reasonable effort will be made to contact the relevant person/people at each university and for each discipline. For example, we will make a second attempt to contact people who have not responded to previous enquiries when we have not been able to contact anyone for that discipline at that university.

Variables

Measured variables
The measured variables include responses to our survey questions.

Analysis Plan

The majority of analysis is descriptive (one exception noted, below). We provide our general approach to analyses.

Our analyses focus on among-(sub-)discipline comparisons and our analyses do not permit comparisons (e.g., ranking) among universities (all data are anonymised for presentation). For each multiple-choice question, we will present results from three perspectives: (1) provide means (or probabilities) with 95% confidence intervals (CI's) at the level of the four disciplines; (2) provide means (or probabilities) with 95% CI's (CI's) for sub-disciplines (to describe trends within Disciplines); (3) provide means (or probabilities) with 95% CI's to compare Biology programmes (sub-disciplines) that included ecology and/or evolution vs. Other sub-disciplines within biological sciences that do not.

We will calculate means and 95% CI's for numerical data (e.g., estimated time spent teaching a subject) using R's lm() and emmeans() functions. We will check assumptions of equal variance and normality by visualizing model residuals. Models will be in the form: **lm(Dep.var ~ Indep.var)**, where Indep.Var is a factor with levels corresponding to Disciplines or sub-disciplines, as appropriate.

We will calculate probabilities and 95% CI's for binary responses (e.g., whether or not a subject is taught) using R's glm() and emmeans() functions. Models will be in the form: **glm(Dep.var ~ Indep.var, family = binomial())**, where Indep.Var is a factor with levels corresponding to Disciplines or sub-disciplines, as appropriate.

We will calculate probabilities and 95% CI's for responses with more than 2 categories (multinomial data; e.g., selecting whether a course teaches statistics from a mathematical perspective, a functional perspective, or both) using R's multinom() function. Models will be in the form **multinom(Dep.var ~ Indep.var); independent variables will be scaled to [0;1] where appropriate, to facilitate the model's fit.** In case of ordered categorical outcomes (e.g. number of credits) we will use an ordinal regression approach, e.g. using the ordinal package in R.

Nesting: As specified in our survey, we assume that responses from different surveys are independent, and therefore do not anticipate nested data. We recognize, however, that the survey may reveal unanticipated non-independence, e.g., shared teaching between disciplines within a university. In this event and if possible, we will analyze the data both with (e.g., including random effects) and without nesting and present both sets of results to clarify whether the approach matters.

Exceptions to this general approach:

- Should the data not meet the assumptions of the analyses described above, we will seek alternative methods. In case of very sparse data or when an appropriate statistical method cannot be found, we will simply describe the data e.g. with frequency histograms.
- Results from multiple-choice questions that allow respondents to "Select all answers that apply" will be reported as separate frequency histograms for each level of (sub-)discipline;
- Question 5a, which describes the length of degree, will be represented in frequency histograms;
- Questions related to time spent teaching (i.e. number of credits) for Study Design and Data Analysis will be analyzed separately for Scottish vs. Non-Scottish UK universities because the lengths of degrees differ between these regions (4 and 3 years, respectively);
- Data describing software used will be presented in frequency histograms.
- We will use two Fisher-Exact tests to determine whether the probability of delivering training in either (a) study design or (b) data analysis differs among disciplines. These analyses will consider responses to the questions, "At your institution, are

undergraduates in your Discipline (Sub-Discipline) taught Study Design at any point in their degree?' and 'At your institution, are undergraduates in your Discipline (Sub-Discipline) taught Data Analysis at any point in their degree?' respectively. We assume responses from separate surveys are independent. We propose to use Fisher-Exact tests because we anticipate that responses will be 100% 'YES' for some disciplines, which can inhibit the fit of logistic regression models.

- We will test whether the probability of returning survey responses differs among Disciplines. Differences could occur, for example, if (a) Disciplines differ in the probability of offering training in study design or analysis and (b) potential survey respondents are less likely to participate in the survey when their programme offers no training. We will use a binomial response model that includes (i) Discipline (factor) and (ii) the total number of contact attempts by our research team to elicit a reply (covariate) as fixed effects and University ID as a random effect. Model: glmer (response (Y/N) ~ Discipline + ContactAttempts + (1|University.ID), family = "binomial")".

Inference criteria

As we aim to characterize training within and among disciplines, our inferences will focus on mean values (with 95% confidence intervals; CI's) for disciplines and mean differences (with 95% CI's) among disciplines.

Data exclusion

In the event that unusual data points compromise model assumptions, we will analyze the data twice (with and without the compromising data) and report both sets of results.

Other

With respect to qualitative data (open responses):

- We will present (in supplementary materials) all open responses, anonymized by identifying respondents with a code and editing responses to remove details that would allow inference of an individual's identity and university.
- We will identify recurrent themes and group open answers by these themes. To do so, one person will group responses by themes and a second person will verify and provide feedback.
- We acknowledge that this approach is arbitrary; by presenting all of the data, however, we allow independent readers to form their own groupings.